

Determinants of 2×2 Matrices

In class, we explored determinants through linear transformations in $\mathbb{R}^2$. The determinant of a matrix represents the **signed area scale factor** between the original region and its transformed image. The sign indicates orientation (whether the transformation preserves or reverses direction), while the magnitude indicates how area changes.

For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is computed by

$$\det(A) = ad - bc.$$

Geometric Interpretation:

- The unit square transforms into a parallelogram.
- $|\det(A)|$ gives the new area.
- $\det(A) > 0$ means orientation is preserved (red image in applet).
- $\det(A) < 0$ means orientation is reversed (green image in applet).

Observations from manipulating individual matrix entries:

- Changing only a stretches or compresses horizontally (rectangle scaling).
- Changing only d stretches or compresses vertically (rectangle scaling).
- Changing only b shears horizontally.
- Changing only c shears vertically.

Shearing changes shape but **does not change area**. So the determinant remains the same when b or c changes alone.

Zero Determinant:

- $\det(A) = 0$ if the image collapses onto a line segment.
- The columns of A become linearly dependent.
- The transformation is not invertible.

$$\det(A) = 0 \iff \text{columns of } A \text{ are linearly dependent} \iff A \text{ is not invertible.}$$

In the applet, this appeared when one arrow overlapped the other or pointed in the exact opposite direction.

Orientation:

$$\det(A) > 0 \Rightarrow \text{orientation preserved} \quad \det(A) < 0 \Rightarrow \text{orientation reversed}$$

These geometric ideas help us understand determinants not only as a formula, but as a meaningful measure of how a matrix transforms the plane.

Determinants and Inverses (for 2×2 matrices):

When a 2×2 matrix is invertible (meaning $\det(A) \neq 0$), we found in The Window Task that

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

The inverse of A can be computed using its determinant:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

You can also reference this formula in your textbook: **Section 2.2, Theorem 4.**

This relationship is special and holds for 2×2 matrices. The situation becomes more complicated for larger matrices, so we must not assume this formula automatically works the same way in higher dimensions.

This formula works only when $\det(A) \neq 0$ because the matrix must be invertible to divide by the determinant.